Electric vehicles might not be the solution for reducing transportation-related environmental impact.

Over the past decade, environmental issues have become the most impactful issues around the globe. Transportation-related environmental impacts are the closest thing that the public can reach and make changes to. Owing to the significant benefits that electric vehicles have and their potential to reduce transportation-related environmental impacts, electric vehicles have become the preferred option when choosing a vehicle. However, this essay will argue that electric vehicles might not be able to reduce transportation-related environmental impact. There are three main reasons why electric vehicles might not be the solution to reduce transportation-related environmental impact. Firstly, the pollution while driving EVs does not go away but is transferred. Secondly, the pollution from making and disposing of an electric vehicle is high. Last but not least, a short battery life cycle would increase the demand for new batteries and worsen the environmental impact brought by manufacturing batteries.

First, compared to internal combustion engine vehicles (ICEVs), EVs

do not emit air pollution when driving, however, the energy of EVs comes from the power grid of the city. According to Ajanovic and Haas (2018), most of the countries heavily rely on coal for electricity generation, in such situation, the carbon-dioxide releases per kWh are high enough to cancel out the greenhouse gas emissions reduce by EVs, the total emissions of greenhouse gases are basically unchanged, the pollution source just move the location, from emitting in the downtown to the power plant. Thus, it is safe to conclude that EVs do not make great contributions to reducing greenhouse gas emissions, resulting in a lack of effectiveness in reducing transportation-related environmental impacts.

Second, the pollution from producing and disposing of EVs is high and toxic. EVs require a larger battery pack. According to Ajanovic and Haas (2018), EVs mostly use lithium-ion batteries (LIBs) as energy storage due to the high energy density and lower cost per kWh, however, the production and disposal of LIBs are extremely harmful to the environment. During the manufacturing of LIBs, according to Rekosuo (2025), the extraction of nickel and lithium has a huge environmental impact in terms of underwater contamination, and a huge

amount of greenhouse gases is emitted when excavating, the manufacturing process of LIBs would also cause releases of toxic substances, inducing acid rain (Kapustina & Izakova, 2023). Apart from that, the energy cost of making an EV is equivalent to burning 10,000 litres of gasoline (Kapustina & Izakova, 2023). The consequences of disposing of LIBs are also worth mentioning, as Jin (2025) mentioned, during the disposal process of LIBs could produce multiple pollution, including heavy metal, organic, dust, fluoride and phosphorus, white, micro-plastic pollution. Those kinds of pollution are often caused by improper handling of waste, lack of recycling infrastructure, and inevitable contamination of soil, water, and air. Those kinds of pollution could cause multiple environmental consequences, including severe neurological effects on animals and, moreover, on humans due to consumption of contaminated foods. Thus, even though EVs do not emit air pollution, the pollution from the production and disposal of EVs cannot be ignored and would cause a huge environmental impact.

Finally, LIBs' usable capacity will decrease over time, as research by Ajanovic and Haas (2018) shows, EVs' LIBs often have a life cycle of 8-10 years or 160,000km, meaning that LIB packs on EVs typically require a

swap after the usable capacity has dropped to 80%. However, according to (IEA, 2016, as cited in Ajanovic and Haas, 2018), the global stock of EVs is rising with increasing slope, thus, the demand for LIB will become much higher in the future. Combined with the swap of batteries and new EVs battery requirements, the demand for new LIBs will increase exponentially. As mentioned before, the production and disposal of LIBs will cause severe environmental impacts, given that swapping LIBs exerts more pressure on LIB disposal facilities. Thus, the increasing rate of LIB manufacturing and disposal leads to increased pollution emissions, which then cause severe environmental problems.

In conclusion, although EVs offer a “zero-pollution” and “green” option compared to ICEVs, the processes behind making EVs are actually harmful to the environment, and the environmental impact caused by those processes might cancel out the benefits brought by EVs. Therefore, it might not be suitable to make EVs the solution to reduce transportation-related environmental issues.

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